Unit: 6 Unit 8: Quadratics
Topic: Solving by Factoring and Other Methods

Name: ________________
Date: ___________

Foundation (Jalapeno)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.
(A) $x = -2, x = -3$
(B) $x = -1, x = -6$
(C) $x = 2, x = 3$
(D) $x = 1, x = 6$
(E) $x = 0, x = 5$

Q2. Factor the expression $x^2 - 7x + 12$.
(A) $(x - 3)(x - 4)$
(B) $(x - 2)(x - 6)$
(C) $(x + 3)(x + 4)$
(D) $(x + 2)(x + 6)$
(E) $(x - 1)(x - 12)$

Q3. Solve the equation $x^2 - 4x - 5 = 0$ by factoring.
(A) $x = 5, x = -1$
(B) $x = 1, x = -5$
(C) $x = 2, x = 3$
(D) $x = -2, x = 5$
(E) $x = 4, x = 1$

Q4. Factor the expression $x^2 - 9x + 20$.
(A) $(x - 4)(x - 5)$
(B) $(x - 2)(x - 10)$
(C) $(x + 4)(x + 5)$
(D) $(x + 2)(x + 10)$
(E) $(x - 1)(x - 20)$

Q5. Solve the equation $x^2 + 2x - 15 = 0$ by factoring.
(A) $x = 5, x = -3$
(B) $x = 1, x = -15$
(C) $x = 2, x = 3$
(D) $x = -2, x = 5$
(E) $x = 4, x = 1$

Q6. Factor the expression $x^2 + x - 12$.
(A) $(x + 4)(x - 3)$
(B) $(x + 2)(x - 6)$
(C) $(x - 4)(x + 3)$
(D) $(x - 2)(x + 6)$
(E) $(x - 1)(x - 12)$

Q7. Solve the equation $x^2 - 2x - 8 = 0$ by factoring.
(A) $x = 4, x = -2$
(B) $x = 1, x = -8$
(C) $x = 2, x = 4$
(D) $x = -2, x = 4$
(E) $x = 5, x = -1$

Q8. Factor the expression $x^2 - x - 6$.
(A) $(x - 3)(x + 2)$
(B) $(x - 2)(x + 3)$
(C) $(x + 3)(x - 2)$
(D) $(x + 2)(x - 3)$
(E) $(x - 1)(x - 6)$

Open Ended Questions (FRQ)

Q9. Solve the equation $x^2 + 4x + 4 = 0$ by factoring and verify the solutions.

Q10. Factor the expression $x^2 - 5x - 14$ and solve the equation $x^2 - 5x - 14 = 0$.

Chef's Tips

- Check for correct factoring
- Verify solutions by plugging them back into the equation
- Use the zero product property to solve the equation